



COOLING TOWER FOUNDATIONS

PURPOSE

This document establishes guidelines for the design of cooling tower foundations.

SCOPE

This document includes the following major sections:

- GENERAL
- DESIGN CONDITIONS
- BASIN LAYOUT CONSIDERATIONS
- BASIN DESIGN
- REINFORCING REQUIREMENTS
- SUMP DESIGN
- HYDRAULIC CONSIDERATIONS
- BUOYANCY
- MAINTENANCE
- JOINTS
- CONSTRUCTION MATERIALS
- REFERENCES
- ATTACHMENTS

APPLICATION

This document should be used for the layout and design of cooling tower foundations or similar basins.

GENERAL

The typical cooling tower foundation will consist of a relatively shallow basin, usually less than 5 feet (1.5m) supporting the tower and a sump pit the depth of which is usually determined by the vertical pumps mounted over it. In colder climates, the pumps are usually horizontal and enclosed within a heated pump house.

This document is intended as a guide in the design of a typical foundation. Additional procedures and methods of analysis may be required for a particular foundation, depending on the job criteria and the judgment of the engineer.



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DESIGN CONDITIONS

Loads

Normally, the cooling tower manufacturer will furnish vertical and horizontal loads, and supply a loading diagram that indicates dry loads (empty), wet loads (operating), and wind or seismic loads for each cooling tower column. The Structural Engineer should review these cooling tower loadings.

Pump Data

Information on pump weights and rated capacity may be obtained from the certified supplier's drawing.

Soil Bearing

Normally, soil bearing pressure is not excessive for the typical cooling tower. Soil bearing should be checked if the foundation is placed on weak soil or fill or whenever a special type of foundation is used. When the structure rests on more than one type of soil, necessary provisions should be made to prevent differential settlement.

Soils Information

Consult the Geotechnical Engineer for recommendations on appropriate lateral earth pressure coefficients, modulus of subgrade reaction, location of the ground water table, allowable soil bearing, depth of frost, and recommended soil preparation. If weak or unstable soil conditions exist, soil preparation under the base slab may be necessary. Sand beds, gravel beds, seal slabs, stabilized materials, or geotextiles may be used to provide a stable surface on which to cast the base slab.

BASIN LAYOUT CONSIDERATIONS

Basin Slab Depth

The bottom of the floor slab should be set below the frost line to prevent heave or buckling.

Tower Support

Typically uplift due to wind load is transferred to the basin through perimeter columns. Anchorage of these perimeter columns is provided by anchor bolts embedded into concrete pedestals adjacent to the basin walls.



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The height of the pedestals is governed by the embedment length of the anchor bolts which is related to the diameter of the bolt. Refer to Structural Engineering Guideline 000.215.STE05121: Anchor Bolt Design Guide, for the anchor bolt length. Anchor bolts are often stainless steel for corrosion resistance.

The interior tower columns are typically supported directly from the basin floor slab without pedestals or anchor bolts. The basin floor slab is designed based on the column size and loads. Columns are usually set about 1 inch (25mm) above the floor and grouted in place.

Solids Retention and Cleaning

A mud sill should be installed across the outlet channel to pump basin. The height of the mud sill is usually between 6 and 12 inches (150 and 300mm). Sufficient height of water should be provided above the mud sill to limit the water velocity at the basin outlet to between 1 and 2 feet per second (0.3 and 0.6m/s). Lower velocity is better from a solids settling standpoint. Setting the top of basin wall 2 feet (600mm) or more above grade will reduce ground dust and trash from entering the basin.

Trash and debris are retained by screens located immediately behind the mud sill in the outlet channel of the basin. Either 3/8 or 1/2 inches (10 or 12mm) mesh screens are commonly used. Stainless steel usually proves to be the best selection; however, galvanized screens have been used successfully. There are 2 factors to be considered for the design of these screens:

- The screens are usually too heavy to remove manually for cleaning. Monorails are often installed over the screens for maintenance purposes.
- Provisions should be made for inserting backup screens or plate behind each screen section so that a screen may be removed and cleaned without shutdown of the basin.

Provisions for cleaning are to drain the basin by use of the distribution pump to pull the level down to the mud sill. The remainder of the water can be pumped from a drain sump with a portable pump. Sludge can be pushed into drain sump and removed with a bucket. Drain lines may be installed if utilization of the distribution pump for drainage is not desired. The disadvantage of this method is that the sludge can often be detrimental to the sewer system. Multicell installations can be partitioned to permit partial shutdown of tower for cleaning and maintenance.

Wall Height

The depth of basin is usually determined by the outlet opening area above the mud sill. A depth of approximately 4 feet (1.2m) is common as this approaches ideal velocities but is willow enough to clean conveniently. Allow 2 inches (50mm) from water level to top of



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perimeter column pedestals, and 6 inches (150mm) minimum freeboard from top of pedestals to top of basin wall.

Water Collection

Waterstops are used on all pipes passing through the basin wall. Stops are usually 1/4 in (6mm) thick steel plate with a minimum edge distance of plate to pipe of about 3 inches (75mm). Sloping the top of the side walls toward the inside of the basin will prevent water ripping off the tower from running down the outside of the basin walls.

Basin Expansion

Expansion can be handled by providing reinforcing ties, keys, and waterstops on the ends of the basin floor and walls. Provision for future expansion is provided by casting holes in the end wall and covering them with plates. The number and size of holes depends on the flow rates required.

BASIN DESIGN

Slab Design

The slab design procedure presented is based on the Westergaard Formula for a concentrated load, P, at the interior of the slab. Minimum slab thickness equals 8 inches (200mm). As an alternative, the elastic plate method for design of mat foundations presented by Winterkorn and Fang may be used to determine the effect of column loads at any point on the mat. (Refer to the References section for additional information.)

$$S = \frac{3(1 + \mu)P}{2\pi h^2} \left(\log_e \frac{L}{a} - 0.6159 \right) \quad \text{Equation 1}$$

where

S = Tensile stress at bottom of slab, psi

μ = Poisson's Ratio for concrete

h = Slab thickness, inches

a = Radius of circle with area equal to actual area of applied load, inches

L = Radius of stiffness, inches



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$$L = \left[\frac{Ech^3}{12(1-\mu^2)K} \right]^{\frac{1}{4}} \quad (\text{Refer to Attachment 01})$$

E_c = Modulus of elasticity for concrete, psi

K = Coefficient of vertical subgrade reaction, psi/inch

Equation 1 has a limited range of application and was modified by Westergaard for more general use. Using a value of $\mu = 0.15$ for Poisson's Ratio, the general formula takes the following form:

$$S = \frac{1.264P}{h^2} \left[\log_{10} \frac{L}{b} + 0.268 \right] \quad \text{Equation 2}$$

where

b = Radius of equivalent distribution of pressure, inches

$b = a$ if a/h is smaller than 0.530 or a/h is greater than 1.724

$b = 1.6 a^2 / h^2$

(Refer to Attachments 3 through 6)

Compute the maximum tensile stress in the slab using the general formula, Equation 2, and the maximum interior column load.

The allowable tensile stress should be taken as the modulus of rupture divided by a factor of safety of 2.

$$S_A = \frac{7.5 \sqrt{f'_c}}{2}$$

Slab reinforcing will be determined from the following criteria:

- Slab bending moment equals either:

M = Tensile stress (Equation 2 modified for strength design) times the section modulus of the uncracked slab

or



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M = Strength design moment transferred to slab by basin wall whichever is greater.

- Required reinforcing: $A_s = \frac{M_u}{a_u d}$
- The reinforcing determined as described above will be placed continuously across the entire slab in a grid pattern.
- Whenever the slab thickness equals or exceeds 10 inches (250mm), 2 layers of reinforcing will be placed in the slab.
- Minimum slab reinforcing will be that required for shrinkage and temperature stresses in accordance with ACI (American Concrete Institute) 318-08, Section 7.12.

The slab will be designed with contraction (control) joints, as detailed in Attachment 06 at 30 foot (10meters) centers maximum.

Wall Design

The basin walls will be designed as vertical cantilevers acted on by the water pressure inside or the earth pressure outside, but not for both pressures acting concurrently.

Reinforced concrete walls will be a minimum of 6 inches (150mm) thick with a single layer of reinforcing according to ACI 318-08. Walls 10 inches (250mm) or thicker will have reinforcement on each face.

Minimum wall reinforcement, vertical and horizontal, will be according to ACI 318-08, Section 14.3 with continuous bars horizontally in the top of the wall and additional bars placed around the perimeter of openings.

Minimum bar size will be #4 (φ12mm) and maximum spacing will be 12 inches (300mm).

Walls will be designed with contraction joints at 30 foot (10meters) centers maximum. Refer to Attachment 05, Figures 1 and 2, and Attachment 06 for recommended joint details. Joints will meet the requirements outlined in ACI 318-08, Section 6.4.

Shear keys and waterstops will be provided at the joint between the slab and walls.

REINFORCING REQUIREMENTS

Refer to Structural Engineering Guideline 000.215.1240: Underground Concrete Pits, for procedures covering reinforcing design and crack control.



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SUMP DESIGN

Follow procedures as outlined in Structural Engineering Guideline 000.215.1240 for the design of cooling tower sumps.

HYDRAULIC CONSIDERATIONS

To prevent turbulence in the pump sump, and thus, prolong the life of the pumps, consider the hydraulic factors given in this procedure, which have been compiled from material by the Peerless Pump Company and standards of HI (Hydraulic Institute).

Pump Sump Dimensions

(Joint effort of Process, Mechanical, and Structural groups)

Suggested sump dimensions for proper pump operation are given in Attachment 04. Enter the graph with the rated capacity of pump and read required dimensions.

The ideal flow to a pump is a straight channel with no obstructions or changes in direction. Water should not flow past 1 pump to reach another.

The bottom of the sump should be flat and level, in the direction of flow, for a minimum of $5\text{-}1/2\text{ D}$ from the centerline of the pump, where D equals the diameter of pump suction bell.

Basin Dimensions

The opening from the basin to the sump channel should be sized for a maximum flow of one foot per second (0.3m/s).

The basin should be sized to provide a minimum of 1 minute capacity for the pumps.

Manufacturer's Approval

Having set the basin and sump dimensions, the engineer should provide a copy of these proposed dimensions for the pump manufacturer's approval.

BUOYANCY

Use a minimum safety factor against flotation of 1.2 (dry condition) for the basin and sump without the tower and pumps. The flotation for the sump and the basin should be checked conservatively, as though they were separate units. Check slab for flexural reinforcement under this condition.



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Wall or slab thickness may be increased or an apron (toe) added to resist buoyancy effects.

Provisions will be made during construction to prevent flotation by means of openings, check valves, French drains, dewatering, or other methods. In flood prone areas, water levels higher than the surface (grade) elevation may occur and should be considered in the design of the basin.

MAINTENANCE

The Structural Engineer should consult with the Equipment Engineer and the cooling tower supplier for recommendations and provisions for proper cleaning and maintenance of the basin.

JOINTS

Cracks in concrete are the result of 3 actions:

- Volumetric change due to drying shrinkage
- Direct stress due to applied loads
- Flexural stress due to bending

Cracks appear where the tensile stress in the concrete exceeds the tensile strength of the concrete itself. Proper sizing and reinforcing of concrete members can eliminate cracks due to applied loads; however, cracks caused by drying shrinkage are inherent in concrete. Although shrinkage cracks in concrete cannot be eliminated, random cracking can be controlled with proper jointing practices.

Joints can be potential points of leakage through cooling tower basins and should be sealed with appropriate joint sealant material to prevent chemicals which could cause corrosion from reaching the reinforcing bars.

For large multibay cooling towers or facilities in cold climates where water may weep through cracks, then freeze, and cause structural damage, consideration should be given to post tensioning long walls to close shrinkage cracks.

Three types of joints used in concrete construction are listed below:

Expansion Joints (Isolation Joints)

Expansion joints allow independent movement, both horizontally and vertically, between 2 masses of concrete. No bond, reinforcing, or mechanical connection exists across the joint. Lateral movement in wall expansion joints may be controlled by keyways of sleeved dowels.



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Contraction Joints (Control Joints)

Contraction joints induce cracking at preselected locations by forming a plane of weakness. Control joints may be saw cut or formed. Load transfer across a control joint is provided by the interlocking of the jagged face formed at the face of the crack or by the use of shear keys or dowels. Reinforcing is normally discontinuous except at saw cut control joints.

Construction Joints

Construction joints provide stopping places during construction. Construction joints are typically located to conform to the jointing pattern and align with control or expansion joints. Load transfer is provided by shear keys or tiebars. Reinforcing is continuous through construction joints.

Refer to Attachment 05, Figures 1 and 2, and Attachment 06 for typical details of expansion joints, construction joints, and control joints.

CONSTRUCTION MATERIALS

Concrete mix for cooling tower foundations will conform to the general requirements outlined in Structural Engineering Guideline 000.215.1240.

Miscellaneous steel, angles, embedded plates, and bolts are typically galvanized; however, certain applications may require stainless steel or fiberglass materials.

Waterstops are typically PVC (Polyvinyl Chloride), meeting the requirements and performance criteria of Corps of Engineers Specification CRD-C572-74. Ribbed type 6 inch (150mm) PVC with a center bulb is the most commonly used waterstop in joints in which movement may occur. Refer to the manufacturer's literature for specific application details.

Cooling tower screens are typically constructed of welded angle or bar with steel woven cloth attached to the perimeter of the frame with counter sunk bolted plate bar. Screens may be galvanized, stainless steel, or specialty coated. Lifting handles should be provided at top for removal of screens for maintenance.

REFERENCES

ACI (American Concrete Institute) 318-08

Corps of Engineers Specification CRD-C572-74

HI (Hydraulic Institute)



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Peerless Pump Company

Fang, Hsai-Yang. Foundation Engineering Handbook, 2nd edition, Van Nostrand Reinhold Co. New York, 1991.

Structural Engineering:

[Guideline 000.215.STE05121](#)

Anchor Bolt Design Guide

[Guideline 000.215.1240](#)

Underground Concrete Pits

ATTACHMENTS

Attachment 01:	Radius of Relative Stiffness Table
Attachment 02:	Coefficient of Subgrade Reaction, K – Table
Attachment 03:	Radius of Equivalent Distribution of Pressure
Attachment 04:	Standards of the Hydraulic Institute Sump Dimensions versus Flow
Attachment 05:	Figure 1: Expansion Joint Details Figure 2: Construction Joint Details
Attachment 06:	Control Joint Details
Attachment 07:	Sample Design 1: Cooling Tower Foundation

Note: *Calculations for Attachment 07 have not been updated to meet the requirements of the latest references and design codes.*

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Radius of Relative Stiffness Table

$$L = \left[\frac{E_c h^3}{12(1 - \mu^2)K} \right]^{\frac{1}{4}}$$

E _c	K	h=3	h=4	h=5	h=6	h=7	h=8	h=9	h=10	h=11	h=12
3,155,000	40	20.6	25.6	30.3	34.7	39.0	43.1	47.1	50.9	54.7	58.4
	50	19.5	24.2	28.6	32.8	36.9	40.7	44.5	48.2	51.7	55.2
	60	18.6	23.1	27.4	31.4	35.2	38.9	42.5	46.0	49.4	52.8
	100	16.4	20.4	24.1	27.6	31.0	34.2	37.4	40.5	43.5	46.4
	130	15.4	19.1	22.6	25.9	29.0	32.1	35.0	37.9	40.7	43.5
	150	14.8	18.4	21.7	24.9	28.0	30.9	33.8	36.6	39.3	41.9
	180	14.2	17.6	20.8	23.8	26.7	29.6	32.3	34.9	37.5	40.1
	200	13.8	17.1	20.2	23.2	26.1	28.8	31.5	34.0	36.6	39.0
	250	13.0	16.2	19.1	21.9	24.6	27.2	29.7	32.2	34.6	36.9
	300	12.5	15.5	18.3	21.0	23.5	26.0	28.4	30.8	33.0	35.3
	400	11.6	14.4	17.0	19.5	21.9	24.2	26.4	28.6	30.7	32.8
	500	11.0	13.6	16.1	18.5	20.7	22.9	25.0	27.1	29.1	31.0

Poisson's Ratio: $\mu = 0.15$

Note: If coefficient of subgrade reaction is not given, use value from the table in Attachment 2.

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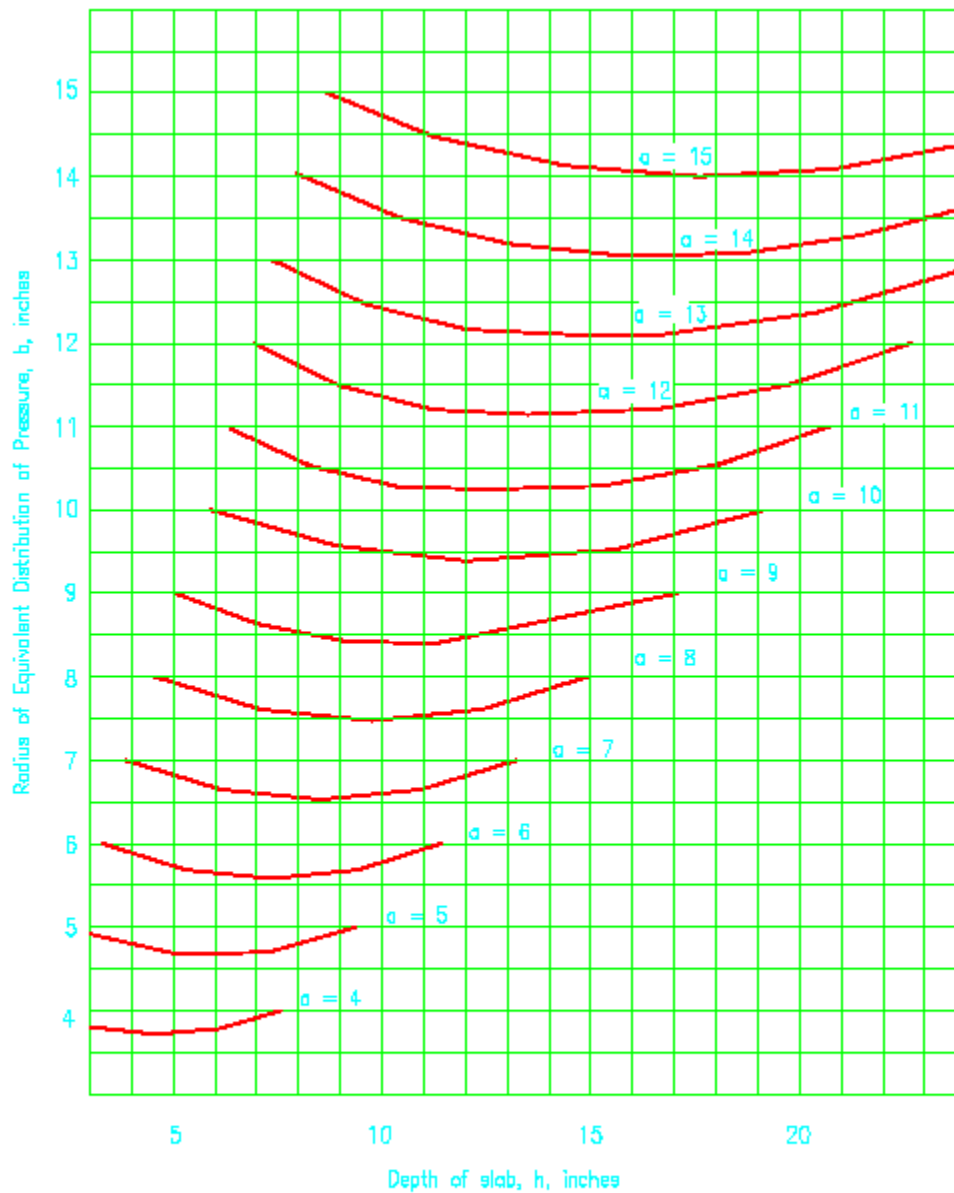
Coefficient of Subgrade Reaction, K-Table

Major Divisions	Soil Groups & Typical Description	Group Symbols	K-Value PCI
Gravel and gravelly soils	Well graded gravel and gravel sand mixtures. Little or no fines.	GW	500
	Well graded gravel sand clay mixtures. Excellent binder	GC	400
	Poorly graded gravel and gravel sand mixtures. Little or no fines.	GP	300
	Gravel with fines, very silty gravel, clayey gravel, and poorly graded gravel sand clay mixtures.	GF	250
Sands and sandy soils	Well graded sands and gravelly sands. Little or no fines.	SW	250
	Well graded sand clay mixtures. Excellent binder.	SC	250
	Poorly graded sands. Little or no fines.	SP	200
	Sand with fines, very silty sands, clayey sands, and poorly graded sand clay mixtures.	SF	180
Fine grained soils having low to medium compressibility	Silts (inorganic) and very fine sands, rock flour, and silty or clayey fine sands with slight plasticity.	ML	150
	Clays (inorganic) of low to medium plasticity, sandy clays, silty clays, and lean clays.	CL	130
	Organic silts and organic silt clays of low plasticity.	OL	100
Fine grained soils having high compressibility	Micaceous or diatomaceous fine sandy and silty soils, and elastic silts.	MH	60
	Clays (inorganic) of high plasticity and fat clays.	CH	50
	Organic clays of medium to high plasticity	OH	40

Note: The values in this table should be verified by the Geotechnical consultant prior to use in design.

COOLING TOWER FOUNDATIONS

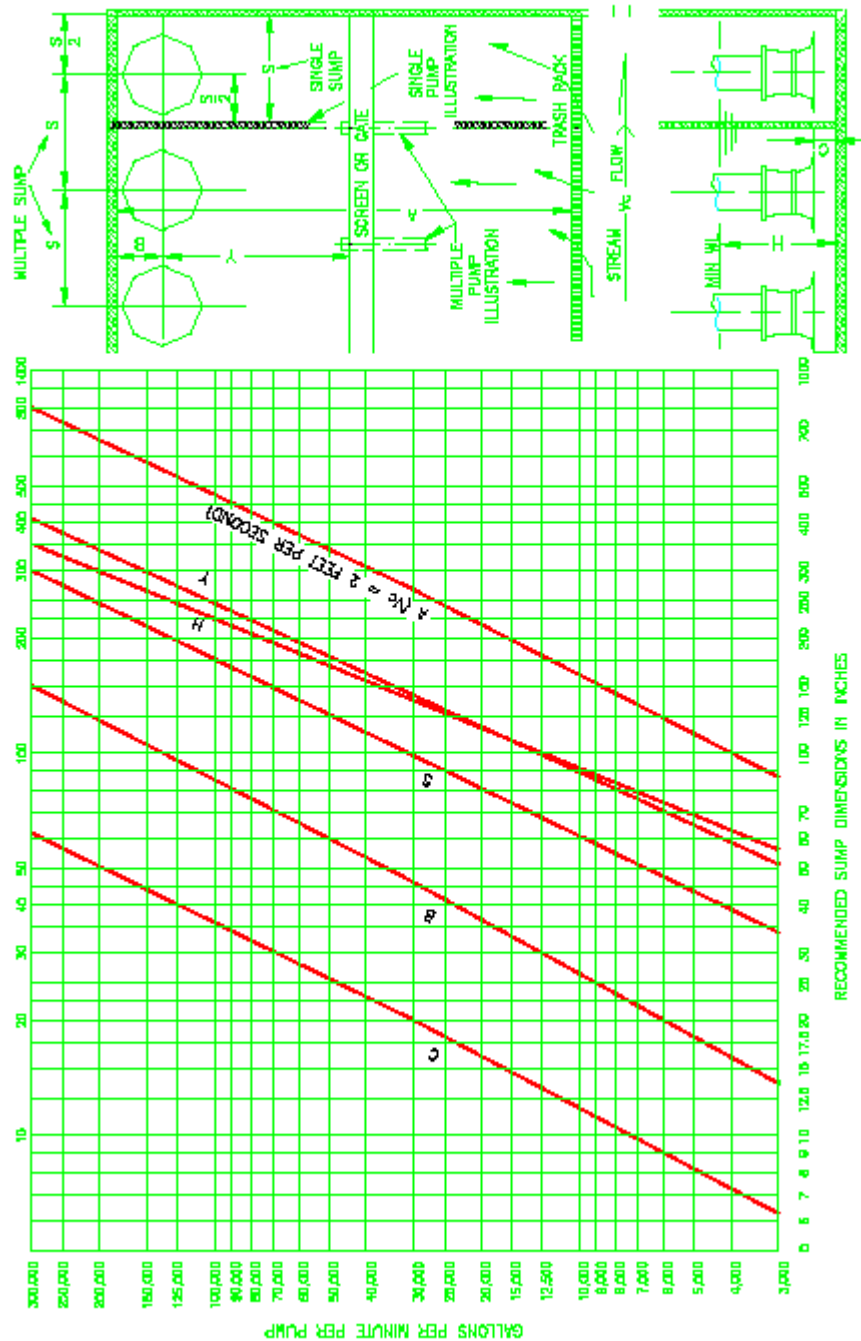
Radius of Equivalent Distribution of Pressure



a = Radius of circle with area equal to actual area of applied load, inches
 $b = a$ when $\frac{a}{h} < 0.530$ or $\frac{a}{h} > 1.724$, $b = \sqrt{1.6 a^2 + h^2} - 0.675$ when $0.53 < \frac{a}{h} < 1.724$

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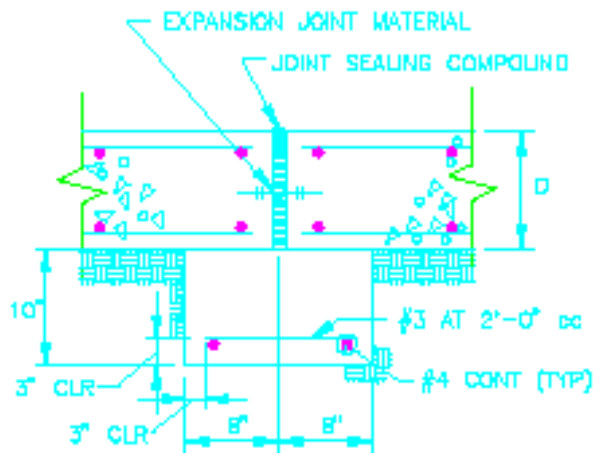
Standards of the Hydraulic Institute Sump Dimensions versus Flow



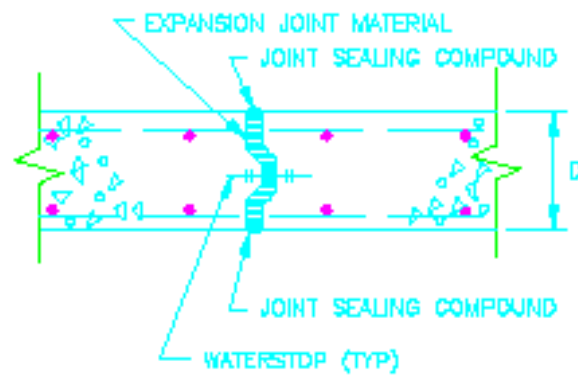
COOLING TOWER FOUNDATIONS

Figure 1: Expansion Joint Details

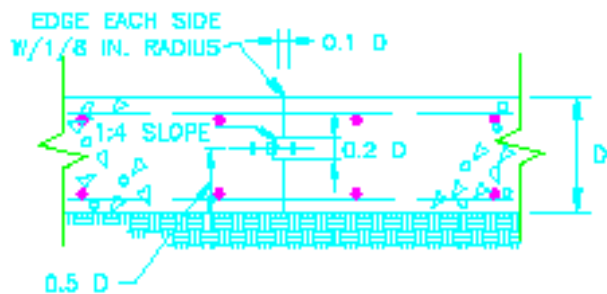
Figure 2: Construction Joint Details



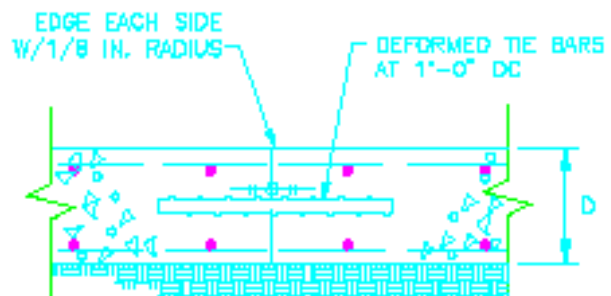
EXPANSION JOINT AT SLAB



EXPANSION JOINT AT WALL



TONGUE AND GROOVE CONSTRUCTION JOINT

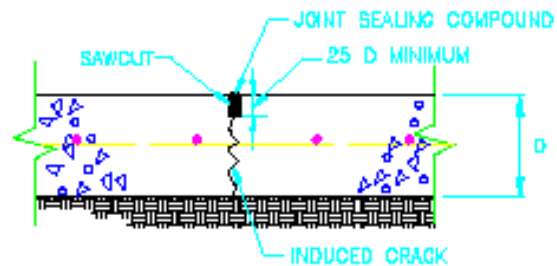


BUTT-TYPE CONSTRUCTION JOINT WITH TIE BARS

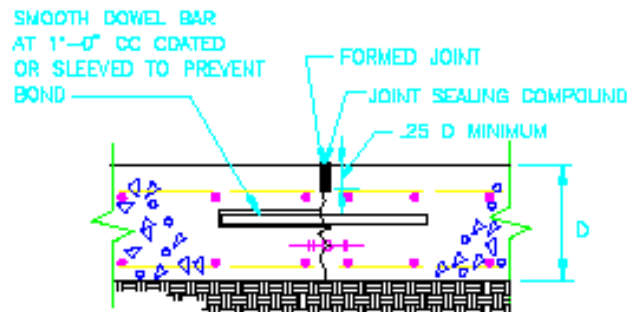
COOLING TOWER FOUNDATIONS

Control Joint Details

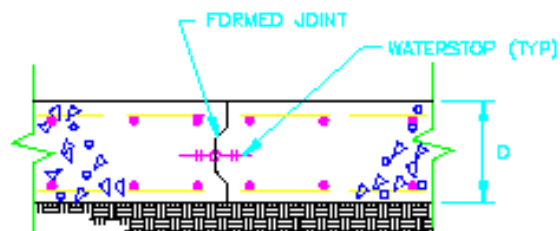
(Wall or Slab)



SAWED CONTROL JOINT



CONTROL JOINT WITH DOWELS

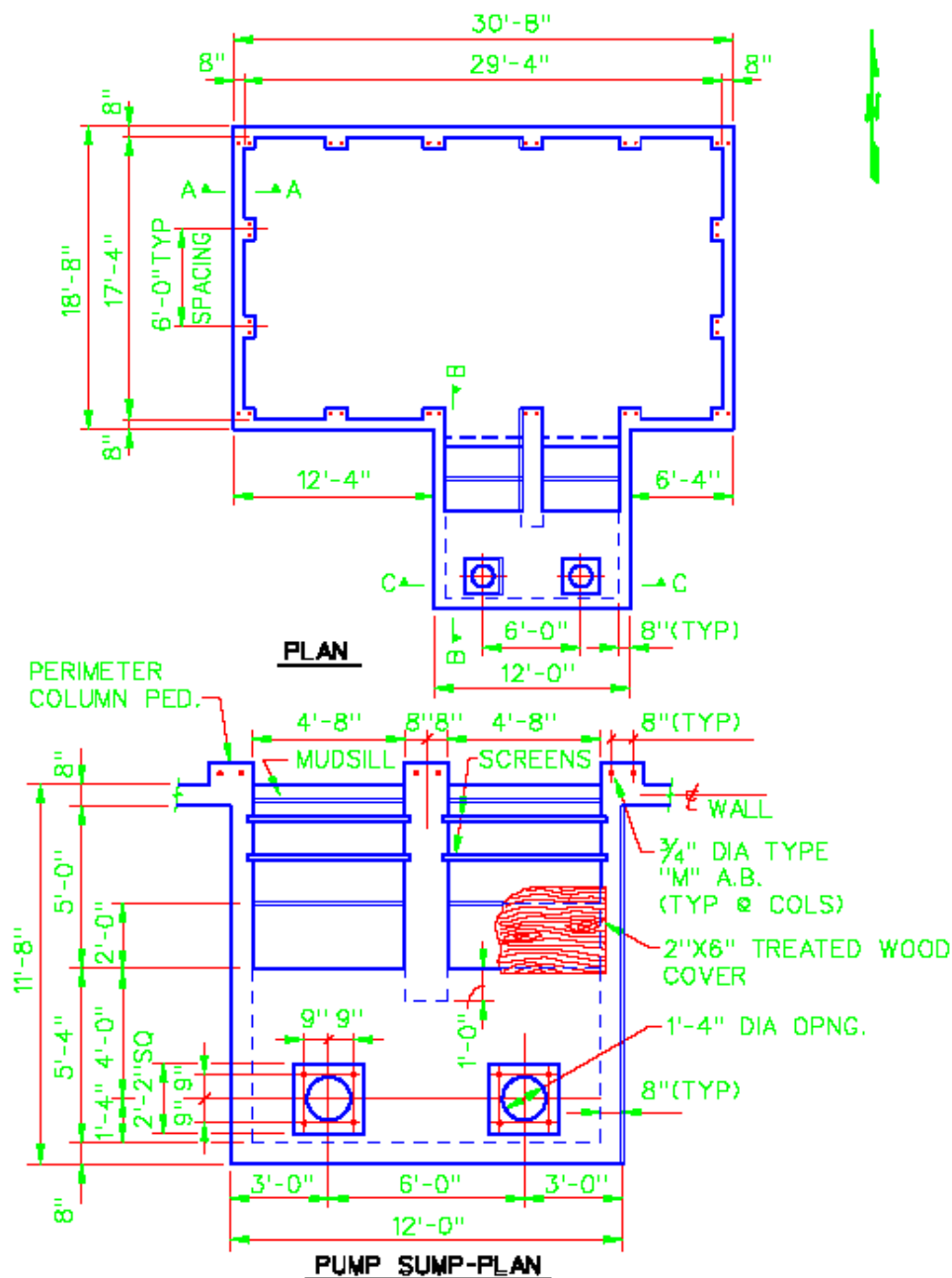


TONGUE AND GROOVE CONTROL JOINT

COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

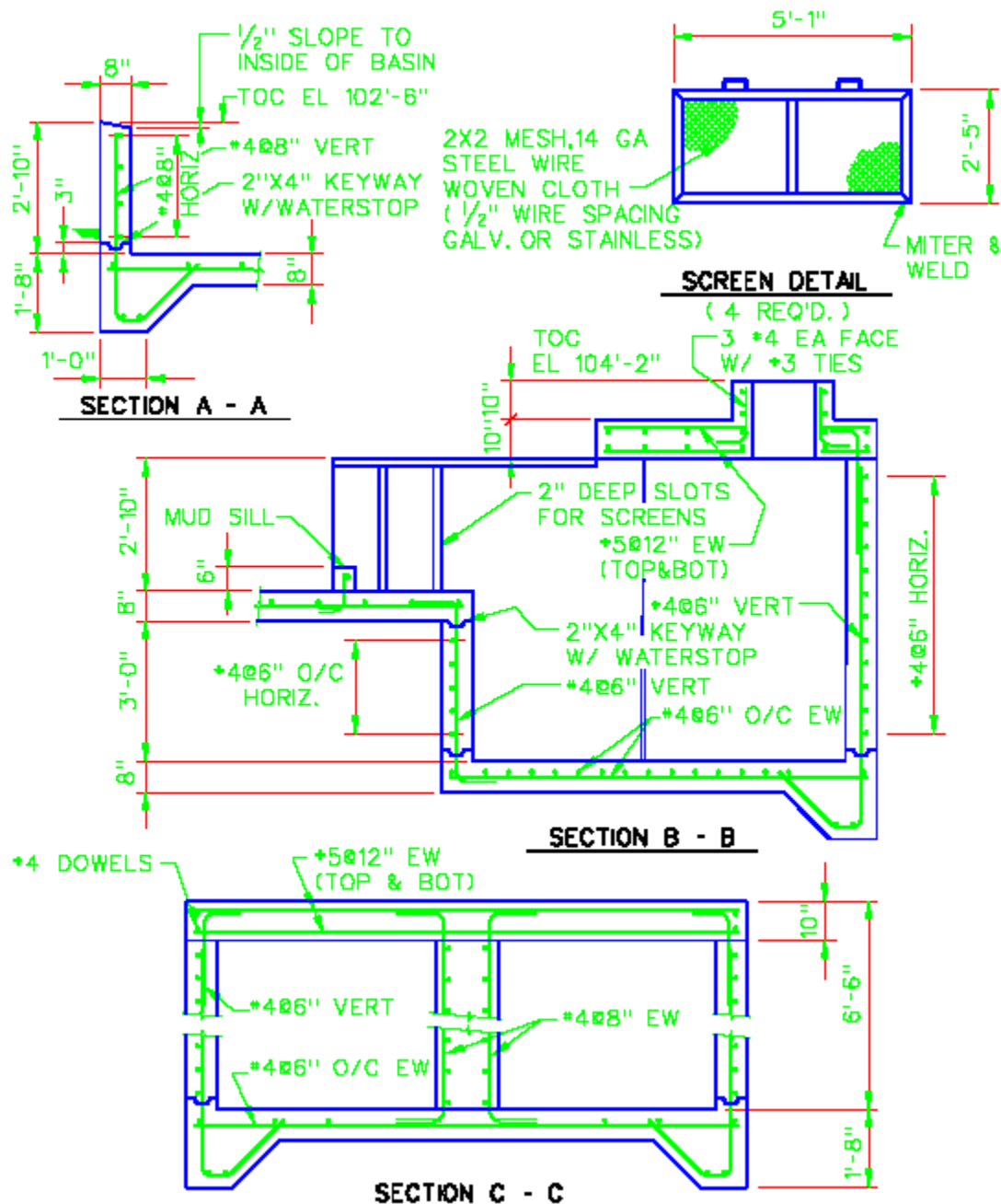
Note: Calculations for Attachment 07 have not been updated to meet the requirements of the latest references and design codes.





COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation



COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

DESIGN DATA

ACI 318-89 Used

Concrete : $f'_c = 3000$ PSI

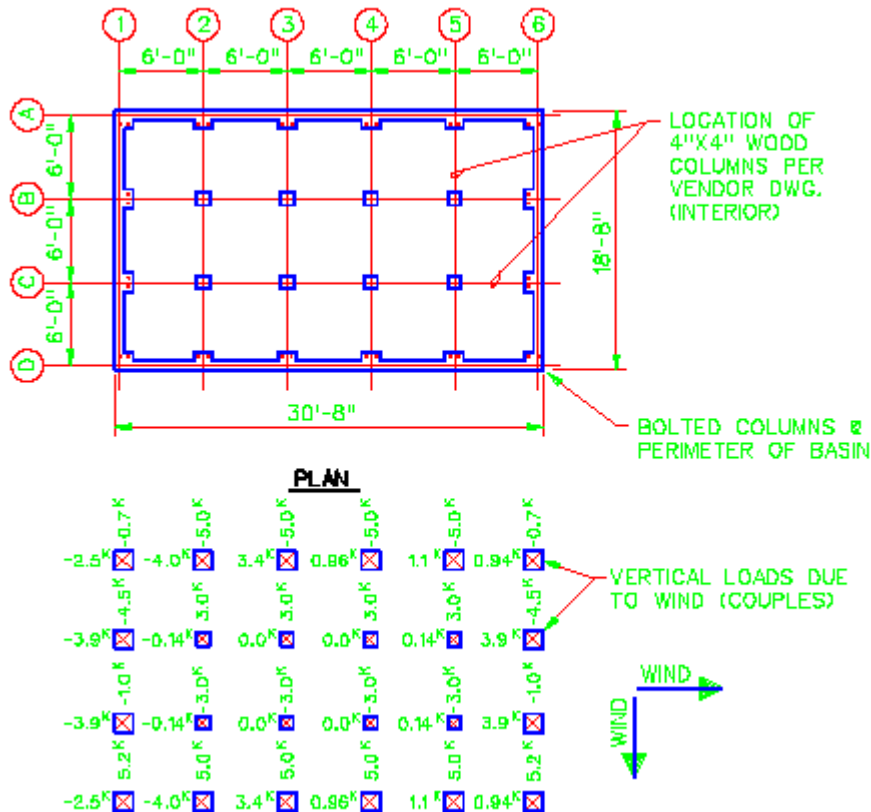
Reinforcing : $f_y = 60,000$ PSI

Earth:

Unit weight = 100 PCF

Allowable bearing (Net) = 3.0 KSF (@ 1'-0" depth)

May increase by 1/3 for wind or seismic



COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

DESIGN DATA (CONTINUED)

Column dead loads:	Operating	Dry
Perimeter (column lines 1, 6, A, & D)	4.2K	2.5K
Interior (column lines B2 to B5, C2 to C5)	4.0K	2.4K

Maximum shear on columns due to wind = 2.0K (occurs @ perimeter)

Vessel Data: From vendor outline drawing

Cooling tower Oper. WT. =	110K
Cooling tower Dry WT. =	65K
Pump capacity =	3,000 GPM
Pump weights =	4.0K Each (2 total)

BASIN WALL DESIGN:

Required height for one minute of pump capacity @ 3,000 GPM

Volume required = $3,000 \text{ GPM} / 7.48 \text{ GAL/FT}^3 = 401 \text{ FT}^3$

Area of basin = $30' \times 18' = 540 \text{ FT}^2$

Minimum depth = $401 \text{ FT}^3 / 540 \text{ FT}^2 = .75 \text{ FT} = 9 \text{ IN}$

Wall height set by vendor requirements : 2'-4" max. water depth

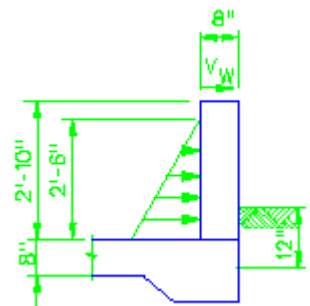
Top of concrete = 102'-6"

Top of basin slab = 99'-8"

2'-10" height w/ 6" freeboard

Determine required reinforcing: (assume wall thickness = 8")

Design wall as a cantilever w/ internal hydrostatic pressure and wind shear from perimeter columns anchored at top of wall.
Neglect axial load.



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Sample Design 1: Cooling Tower Foundation

$$M_{\text{water}} = 62.4(2.5)^3/6.0 = 162.5 \text{ LB-FT/FT} = 0.163 \text{ K-FT/FT}$$

$$M_{\text{wind}} = 2.0K (2.83 \text{ FT})/6.0 \text{ FT} = 0.94 \text{ K-FT/FT}$$

Note: For this example assume wind shear distributed over 6 FT of wall.

BASIN WALL DESIGN (CONTINUED):

Could design perimeter column pedestals to resist entire wind shear.

$$M_u = .75(1.4(0.163 \text{ K-FT}) + 1.7(0.94 \text{ K-FT})) = 1.4 \text{ K-FT/FT wall}$$

$$F = bd^2/12,000 = 12(4)^2/12,000 = .016$$

$$K_u = M_u/F = 1.4/0.016 = 88 ; a_u = 4.42$$

$$A_s = M_u/a_u d = 1.4/[4.42(4)] = 0.08 \text{ IN}^2/\text{FT} \quad (4/3)A_s = (4/3)0.08 = 0.11 \text{ IN}^2/\text{FT}$$

Min. reinforcement per ACI 318-89, section 14.3

$$A_{sv} = .0012(12)(8) = 0.12 \text{ IN}^2/\text{FT}$$

$$A_{sh} = .0025(12)(8) = 0.24 \text{ IN}^2/\text{FT}$$

Use : #4 @ 8" O/C (horizontal & vertical) ($A_s = 0.30 \text{ IN}^2$)

Crack control : (ACI 318-89, section 10.6)

$$Z = f_s (d_c A)^{1/3} < 145 \text{ K/IN} \quad f_s = 0.6(60 \text{ KSI}) = 36 \text{ KSI} \quad A = 2 d_c S = 2(2)(8) = 32$$

$$d_c = 4" \text{ use } 2" \text{ max.}$$

$$Z = 36(2(32))^{1/3} = 144 \text{ K/IN} \quad \text{ok}$$

BASIN SLAB DESIGN

Assume thickness = 8"

(wind page 3)

$$\text{Max. interior column load} = \underbrace{4.0K} + \overbrace{3.0K} = 7.0K$$

(oper. page 4)

COOLING TOWER FOUNDATIONS**Sample Design 1: Cooling Tower Foundation**

Area of applied load = 4" X 4" = 16 IN²

$a = (16/\pi)^{1/2} = 2.26 \text{ IN}$; $a/h = (2.26)/8 = 0.283 < 0.53$ therefore, $b=a$ (Figure 1)

For $K = 100$ with $h = 8"$ from Table 1 : Radius of relative stiffness, $L = 34.2 \text{ IN}$

Compute slab tensile stress:

$S = (1.264P/h^2)(\log_{10} (L/b) + .268)$; $P = 7,000 \text{ LB}$ from above

$$S = \frac{1.264(7,000)}{8^2} \left(\log_{10} \frac{32.4}{2.26} + .268 \right) = 200 \text{ PSI}$$

$$S_{\text{allow}} = 7.5 / 2 \sqrt{f'_c} = 3.75 \sqrt{3,000} = 205 \text{ PSI} > 200 \text{ PSI} \quad \text{ok}$$

BASIN SLAB DESIGN (CONTINUED):

Compute reinforcing required due to column load:

$$P_u = 0.75(1.4(4.0K) + 1.7(3.0K)) = 8.03K$$

$$S = \frac{1.264(8,030)}{8^2} \left(\log_{10} \left(\frac{34.2}{2.26} \right) + .268 \right) = 229.6 \text{ PSI}$$

$$\text{Section modulus} = bd^2/6 = 12(8)^2/6 = 128 \text{ IN}^3$$

$$M_u = S_u bd^2/6 = 229.6(128)/12,000 = 2.45 \text{ K-FT/FT slab}$$

$$F = bd^2/12,000 = 12(4)^2/12,000 = 0.016$$

$$K_u = M_u/F = 2.45/0.016 = 153 ; a_u = 4.34$$

$$A_s = M_u/a_u d = 2.45/4.34(4) = 0.14 \text{ IN}^2/\text{FT} ; (4/3)A_s = (4/3)0.14 = 0.19 \text{ IN}^2/\text{FT}$$

$$A_{s\text{min}} = 0.0033(12)(4) = 0.16 \text{ IN}^2/\text{FT}$$

$$\text{Minimum temperature reinforcing} = 0.0018(8)(12) = 0.17 \text{ IN}^2/\text{FT} \\ [\text{ACI 318-89, section 7.12}]$$

Use #4 @ 10" O/C E.W. ($A_s = 0.24 \text{ IN}^2$)

Check basin slab reinforcing for moment @ base of wall:

COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

$$M_{\text{wind}} = 2.0K (2.83' + 4''/12) = 6.33 \text{ K-FT}/6 \text{ FT section} = 1.1 \text{ K-FT/FT}$$

$$M_{\text{water}} = \frac{0.0624(2.5)^2}{2} \left(\frac{2.5}{3} + \frac{4''}{12} \right) = 0.23 \text{ K-FT/FT} \quad (\text{PG 4})$$

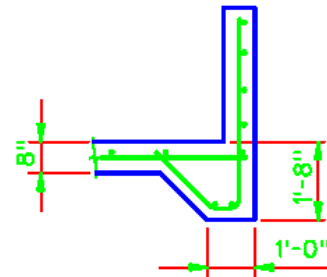
$$M_u = .75[1.4(.23) + 1.7(1.1)] = 1.64 \text{ K-FT/FT} < 2.45 \text{ K-FT/FT} \quad (\text{PG 4})$$

Therefore, reinforcing previously designed is adequate

DEVELOPMENT OF VERTICAL WALL REINFORCING:

A thickened slab may be used along the perimeter of the basin directly below the vertical walls to permit full development of vertical wall reinforcing.

$$l_{dh} = \frac{1200(0.5)}{\sqrt{3,000}} = 11'' + 3'' \text{ CLR} = 14'' \quad \text{ok}$$



CHECK BUOYANCY OF BASIN:

$$\text{Basin weight: Walls} = 2(30.67 + 17.33)(8/12)(2.83)(0.15) = 27.3K$$

$$\text{Slab} = (30.67)(18.67)(8/12)(0.15) = 57.5K$$

$$\text{Thick slab} = 2(30.67 + 17.33)(1.17)(0.15) = 16.8K$$

$$\text{Total} = 101.6K$$

Buoyant force: (assume water table @ surface)

COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

$$F_b = [(30.67)(18.67)(1.0) + 2(30.67 + 18.0)(1.17)](0.0624) = 42.8K$$

$$F.S. = 101.6K/42.8K = 2.3 > 1.2 \quad \text{ok} \quad \boxed{\text{Therefore, basin will not float}}$$

CHECK SOIL BEARING:

$$\text{basin area} = 30.67(18.67) = 572.6 \text{ FT}^2$$

$$\text{Tower Oper. WT.} = 110K$$

$$\text{Basin WT.} = 101.6K$$

$$\text{Water WT.} = 29.33(17.33)(2.5)(0.0624) = \underline{79.3K}$$

$$\text{Total} = 290.9K$$

$$S.B. = 290.9K/572.6 \text{ FT}^2 = 0.51 \text{ KSF} \ll \text{allowable} = 3 \text{ KSF} \quad \text{ok}$$

CHECK BASIN SLAB REINFORCING FOR UPLIFT FORCE:

Since basin is bottomed only 1'-0" below grade, the uplift force caused by the water table being at the surface is small, therefore it is not critical.

ANCHOR BOLT DESIGN:

Note: For brevity, anchor bolt design will not be shown here. See technical practice 000.215.1207, "Anchor Bolt Design Criteria" for anchor bolt design.

PUMP SUMP DESIGN:

Pump data: From vendor outline drawing (2 pumps total)

WT. = 4.0K ea. Baseplates = 2'-0" X 2'-0" X 1" thick

Capacity = 3,000 GPM Suction Bell = 1'-1" Diameter

RECOMMENDED SUMP DIMENSIONS: (Figure 3)

Minimum water level: H = 58" use: 5'-0"

Pump spacing: S = 34" use: 3'-0"

Bell clearance: C = 6" use: 0'-6"

COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

Centerline pump to back wall: $B = 13''$ use: 1'-3"

Centerline pump to screen: $Y = 52''$ use: 4'-6"

Centerline pump to change in elevation of sump bottom = $5 \frac{1}{2} D = 5 \frac{1}{2}(1'-1'') = 6'-0''$

DESIGN PUMP SUPPORT SLAB: (assume 10" thickness) $d = 7.5''$

Assume slab simply supported; width of effective section = $2'-2'' + 3(10'') = 4'-8''$ strip
(ped. width) (slab thk.)

$$w = .83(0.15)(4.67') = 0.58 \text{ K/FT}$$

$$R = 4.0\text{K} + 1/2(10')(0.58 \text{ K/FT}) = 6.9\text{K}$$

$$M_{\max.} = wl^2/8 + Pa = 0.58(10')^2/8 + 4.0(2') = 15.25 \text{ K-FT}/4.67 \text{ FT}$$

$$= 3.3 \text{ K-FT/FT}$$

$$M_u = 1.4(3.3 \text{ K-FT/FT}) = 4.6 \text{ K-FT/FT}$$

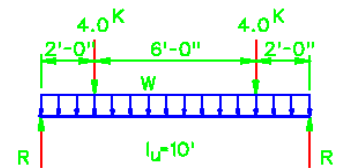
$$F = bd^2/12,000 = 12(7.5)^2/12,000 = 0.056 ; K_u = 4.6/0.056 = 82 ;$$

$$a_u = 4.42 ; A_s = M_u/a_u d = 4.6/4.42(7.5) = 0.14 \text{ IN}^2/\text{FT} ;$$

$$(4/3)A_s = (4/3)0.14 = 0.19 \text{ IN}^2/\text{FT}$$

$$A_{s\min.} = 0.0033(12)(7.5) = 0.30 \text{ IN}^2/\text{FT}$$

Use #5 @ 12" O/C E.W. (top & bottom) ($A_s = 0.31 \text{ IN}^2/\text{FT}$)



SUMP WALL DESIGN: (assume wall thickness = 8")

Design wall as a cantilever beam for internal hydrostatic pressure.

$$M_{\text{water}} = 0.0624(6.17)^3/6 = 2.44 \text{ K-FT/FT}$$

$$M_u = 1.4(2.44) = 3.42 \text{ K-FT/F}$$

$$F = bd^2/12,000 = 12(4)^2/12,000 = 0.016$$

COOLING TOWER FOUNDATIONS

Sample Design 1: Cooling Tower Foundation

$$K_u = M_u/F = 3.42/0.016 = 214 ; a_u = 4.28$$

$$A_s = M_u/a_u d = 3.42/4.28(4) = 0.20 \text{ IN}^2/\text{FT}$$

SUMP WALL DESIGN (CONTINUED):

$$(4/3)A_s = 4/3(0.2) = 0.27 \text{ IN}^2/\text{FT}$$

Minimum reinforcement per ACI 318-89, section 14.3

$$A_{SV} = 0.0012(12)(8) = 0.012 \text{ IN}^2/\text{FT}$$

$$A_{SH} = 0.0025(12)(8) = 0.24 \text{ IN}^2/\text{FT}$$

Use #4 @ 8" O/C (horizontal & vertical) ($A_s = 0.30 \text{ IN}^2$)

Crack control:

$$Z = f_s (d_c A)^{1/3} < 145 \text{ K/IN } f_s = 0.6(60 \text{ KSI}) = 36 \text{ KSI } A =$$

$$2d_c S = 2(2)(8) = 32$$

$$d_c = 4" \text{ use } 2" \text{ max.}$$

$$Z = 36(2(32))^{1/3} = 144 \text{ K/IN } \quad \text{ok}$$

SUMP BASE SLAB:

(assume thickness = 8")

Slab must be able to resist moment from base of walls.

Use #4 @ 8" O/C E.W. as a minimum

Check base slab for uplift: (4'-8" bury) assume uniformly distributed simple span; $l_u = 10'$

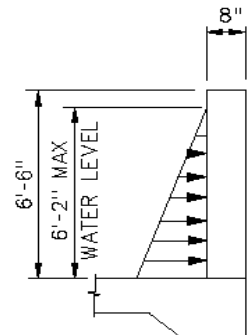
$$W_{\text{uplift}} = 0.0624(4.67)(1.0) = 0.29 \text{ K/FT } \quad \uparrow$$

$$W_{\text{D.L.}} = 0.83(0.15)(1.0) = 0.12 \text{ K/FT } \quad \downarrow$$

$$M_{\text{max.}} = (0.29 - 0.12)(10)^2/8 = 2.13 \text{ K-FT/FT}$$

$$M_u = 1.4(2.13) = 3.0 \text{ K-FT/FT} < 3.42 \text{ K-FT (PG 8) from wall}$$

Use #4 @ 8" O/C E.W. ($A_s = 0.30 \text{ IN}^2$)



COOLING TOWER FOUNDATIONS**Sample Design 1: Cooling Tower Foundation****CHECK BUOYANCY OF SUMP:** (assume water table @ surface)

$$\text{Sump WT.: Walls} = [2(10') + 12'](8/12)(6.5)(0.15) = 20.8\text{K}$$

$$\text{Slab} = 12'(11')(8/12)(0.15) = 13.2\text{K}$$

$$\text{Top slab} = 12'(6')(10/12)(0.15) = 9.0\text{K}$$

$$\text{Inter. walls} = 3'(10')(8/12)(0.15) + [2.83(3') + 6.5(3')](1.33)(0.15) = \underline{8.6\text{K}}$$

$$\text{Total} = 51.6\text{K}$$

CHECK BUOYANCY OF SUMP: (CONTINUED)

$$\text{Buoyant force: } F_b = 4.67'(11')(12)(0.0624) = 38.5\text{K}$$

$$\text{F.S.} = 51.6\text{K}/38.5\text{K} = 1.34 > 1.2 \quad \text{ok}$$

sump will not float

CHECK SOIL BEARING:

Soil bearing @ sump ok, since displaced soil weight is greater than full basin weight.

CHECK VELOCITY @ BASIN OPENING: (flow w/ one pump = 3,000 GPM)

Depth of water = 2'-0" (normal operation)

Clear width of opening = 10'-0"

$$\text{Velocity} = \frac{3000}{7.48(2)(10)(60)} = 0.34 \text{ FT/SEC} < 1.0 \quad \text{ok}$$

Note: Calculations for this attachment have not been updated to meet the requirements of the latest references and design codes.